Skellefteå 2021-08-12

Project: 84690 Delhi Fire Station

To whom it may concern,

Door type	:VL3110
Dimensions	:CWxCH 6000x5400

We certify that the above calculated door have been designed to withstand a differential pressure of 1.15kPa x beam safety factor of 1,5.

The doors should be installed and serviced by authorized installers certified by ASSA ABLOY

Best regards



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Oskar Vestermark

Technical Manager

ASSA ABLOY ENTRANCE SYSTEMS

MEGADOOR

ASSA ABLOY

Calculation of uniformly loaded intermediate beam

Project: **Delhi Fire Station**
Project number: 84690
Date: **2021-08-12**
Name: Mats Brännström

Nominal differential pressure: $P := 1150 \cdot \frac{N}{m^2}$

Fabric contribution: $cf := 1.0$

Reduction factor due to profile rotation: $rf := 0.970$

Actual pressure to be supported by intermediate beam: $p := \frac{P}{cf}$ $p = 1150 \ \frac{N}{m^2}$

Moment of inertia: $I := rf \cdot 1170000 \cdot mm^4$ $I = 1134900 \ mm^4$

Support distance: $l := 6000 \cdot mm$

Plastic section modulus: $Z := rf \cdot 27000 \cdot mm^3$ $Z = 26190 \ mm^3$

Yield point: $\sigma_y := 240 \cdot \frac{N}{mm^2}$

Modulus of elasticity: $E := 70000 \cdot \frac{N}{mm^2}$

Safety factor acc. to EN 12604: $n_s := 1.5$

Allowable stress: $\sigma_a := \frac{\sigma_y}{n_s}$ $\sigma_a = 160 \ \frac{N}{mm^2}$

Max allowed beam spacing: $d. := \frac{\sigma_a \cdot 8 \cdot Z}{l^2 \cdot p}$ $d. = 809.7 \ mm$

Actual spacing: $d := 706 \cdot mm$

Beam stress at actual spacing: $\sigma := \frac{d}{\left(\frac{8 \cdot Z}{l^2 \cdot p}\right)}$ $\sigma = 140 \ \frac{N}{mm^2}$

Skenbar förkortning
Apparent shortening
Contraction apparente

$$\Delta := \int_{0 \cdot mm}^{l} \sqrt{1 + \left(\frac{d}{dx}\left(\frac{5 \cdot d \cdot l \cdot p \cdot l^3}{384 \cdot E \cdot I} \cdot \left(1 - \frac{4 \cdot x^2}{l^2}\right) \cdot \left(1 - \frac{4 \cdot x^2}{5 \cdot l^2}\right)\right)\right)^2} \, dx - l$$

$\Delta = 12.308 \ mm$

Deflection: $\delta := \frac{5 \cdot d \cdot p \cdot l^4}{384 \cdot E \cdot I}$ $\delta = 172 \ mm$